



Barker
College

Student Number

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Friday 17th May, 2024

50 minutes

225 copies

Year 11 MATHEMATICS ADVANCED Assessment Task 2

Topics				
Algebra, Surds and Number	Functions and Graphs	Transformations and Symmetry	Trigonometry	The Coordinate Plane

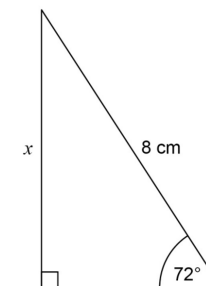
General Instructions	Sections	Marks
	Question 1	/ 11
	Question 2	/ 9
	Question 3	/ 10
	Question 4	/ 12
	Question 5	/ 8
	Total Marks:	/ 50

- Answer in the spaces provided.
- Write using blue or black pen. No pencil.
- Show **all necessary working** to gain full marks.
- Marks may not be awarded for careless or poorly arranged working.
- Diagrams are **not to scale**.
- Approved calculators may be used.
- A reference sheet is provided separately.

Question 1 (11 marks)

- a. Use trigonometry to work out the length of x .
Round your answer to **2 decimal places**.

2



- b. Solve $5x^2 = 10x + 4$. Give your answers to **2 decimal places**.

2

- c. Simplify $\frac{4p-12p^2}{3} \times \frac{6p}{3p^2-p}$

2

d. Let $P(x) = x^3 - 3x^2 - 2x + 1$. Find $P\left(-\frac{1}{2}\right)$ 1

e. State the **related acute angle** for the following angles:

α) 300° 1

β) -600° 1

f. Solve $|7x + 15| = 20$ 2

Question 2 (9 marks)

a. Find the angle of inclination for the graph of $y = 3x + 5$.
Round your answer to **the nearest minute**. 1

b. State the **domain** of the function $f(x) = \frac{1}{\sqrt{8-x}}$ 1

c. Determine, **showing all working**, whether the function $f(x) = \frac{3x}{5x^2-1}$ is even, odd or neither. 2

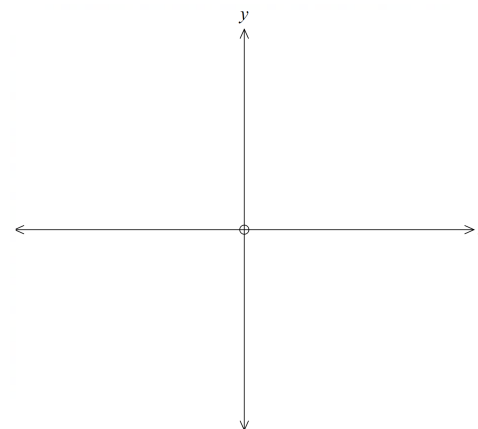
- d. Determine the equation, in **general form**, of the line which is perpendicular to the line joining the points $A(3,5)$ and $B(6,10)$, and passes through the point $C(-3,-1)$. 3

- e. Solve $\cos \theta = \frac{1}{\sqrt{2}}$ for $0^\circ \leq \theta \leq 360^\circ$ 2

Question 3 (10 marks)

- a. Given $f(x) = 5 - x$ and $g(x) = 3x + 7$, **simplify** $f(2x) + g(f(x))$. 2

- b. Sketch the function $y = 2 - 2^{-x}$ on the axes below. Ensure that you show **all key features**. 3



c. If $\sin \theta = \frac{\sqrt{7}}{6}$ and $90^\circ < \theta < 180^\circ$, find the **exact value** of $\tan \theta$. 2

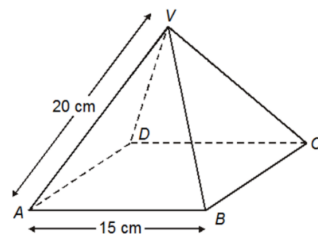
d. Solve $3 \tan^2 \alpha + 4 = 5$ for $0^\circ \leq \alpha \leq 360^\circ$ 3

Question 4 (12 marks)

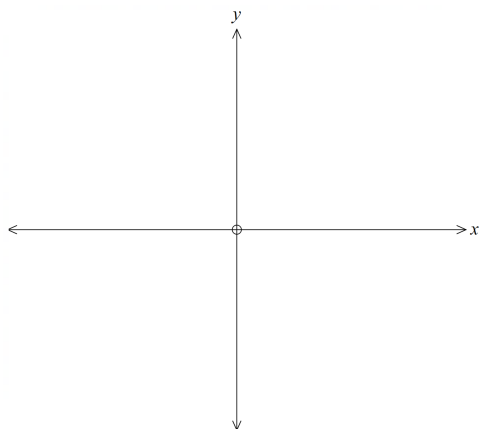
a. A curve has equation $y = 4x^2 + 5x + 3$ 3
 A line has equation $y = x + 2$
 Find the **point(s) of intersection** of the line and the curve.
 Use an algebraic method.

b. A circle has equation $x^2 + y^2 - 10x + 16y - 80 = 0$
 Find the coordinates of the **centre** of the circle and the **radius** of the circle. 3

- c. For the right, square-based pyramid below, calculate angle VAC .
Round your answer to the **nearest degree**. 3



- d. Sketch the function $y = -(x + 1)^3(x - 2)(x - 3)^2$ on the axes below.
Ensure that you show ***x*- and *y*-intercepts**. 3



Question 5 (8 marks)

- a. Simplify $\tan x \cos x \operatorname{cosec} x$ 1

- b. Prove $\frac{1 - \sin^2 x \cos^2 x}{\sin^2 x} = \cot^2 x + \sin^2 x$ 3

- c. Solve $2 \sin^2 x - \cos^2 x - \frac{1}{2} \sin x = \frac{3}{2}$ for $0^\circ \leq x \leq 360^\circ$
Round your answer(s) to the nearest degree.

4

END OF ASSESSMENT

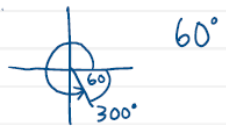
Year

11

Year 11 Assessment Task 2 2024

Page

1

1	(a) $\sin(72^\circ) = \frac{x}{8}$ $x = 8 \sin(72^\circ)$ $= 7.61 \text{ cm}$ (b) $5x^2 = 10x + 4$ $5x^2 - 10x - 4 = 0$ $x = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(5)(-4)}}{2(5)}$ $x = 2.34 \text{ or } x = -0.34$ (c) $\frac{4p-12p^2}{3} \times \frac{6p}{3p^2-p}$ $= \frac{4p(1-3p)}{3} \times \frac{2\cancel{6}p}{\cancel{p}(3p-1)}$ $= \frac{-4p(\cancel{3p-1})}{1} \times \frac{2}{\cancel{3p-1}}$ $= -8p$ (d) $P(-\frac{1}{2}) = (-\frac{1}{2})^2 - 3(-\frac{1}{2})^2 - 2(-\frac{1}{2}) + 1$ $= \frac{9}{8}$ (e) (A)  (B) 	(f) $7x+15 = 20$ $7x+15=20 \text{ or } 7x+15=-20$ $7x=5 \quad 7x=-35$ $x=\frac{5}{7} \quad x=-5$ 2 (a) $y = \underset{\substack{\uparrow \\ m}}{3}x + 5$ $m = \tan \theta$ $3 = \tan \theta$ $\theta = \tan^{-1}(3)$ $\theta = 71^\circ 34'$ (b) $8 - x > 0$ $8 > x$ $x < 8$ (c) $f(x) = \frac{3x}{5x^2-1}$ $f(-x) = \frac{3(-x)}{5(-x)^2-1}$ $= \frac{-3x}{5x^2-1}$ $-f(x) = -\left(\frac{3x}{5x^2-1}\right)$ $= \frac{-3x}{5x^2-1}$ $f(-x) = -f(x)$ $\therefore \text{ODD}$
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$$(d) m_{AB} = \frac{10-5}{6-3} = \frac{5}{3}$$

$$l_{m_{AB}} = -\frac{3}{5}$$

$$\begin{aligned} y - (-1) &= -\frac{3}{5}(x - (-3)) \\ y + 1 &= -\frac{3}{5}(x + 3) \\ 5(y + 1) &= -3(x + 3) \\ 5y + 5 &= -3x - 9 \\ 3x + 5y + 14 &= 0 \end{aligned}$$

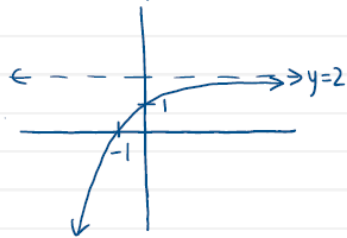
$$(e) \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$



$$\begin{aligned} \theta &= 45^\circ, 360^\circ - 45^\circ \\ \theta &= 45^\circ, 315^\circ \end{aligned}$$

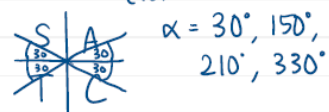
$$\begin{aligned} 3 \quad (a) \quad f(2x) &= 5 - (2x) \\ &= 5 - 2x \\ g(f(x)) &= 3(5 - x) + 7 \\ &= 15 - 3x + 7 \\ &= 22 - 3x \\ f(2x) + g(f(x)) &= 5 - 2x + 22 - 3x \\ &= 27 - 5x \end{aligned}$$

$$\begin{aligned} (b) \quad y\text{-int: Sub } x=0 \\ y &= 2 - 2^{-0} \\ &= 1 \\ x\text{-int: Sub } y=0 \\ 0 &= 2 - 2^{-x} \\ 2^{-x} &= 2^1 \\ -x &= 1 \\ x &= -1 \end{aligned}$$



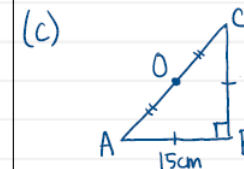
$$\begin{aligned} (c) \quad \sqrt{6^2 - \left(\frac{\sqrt{7}}{2}\right)^2} &= \sqrt{29} \\ \tan \theta &= -\frac{\sqrt{7}}{\sqrt{29}} \end{aligned}$$

$$\begin{aligned} (d) \quad 3 \tan^2 x + 4 &= 5 \\ \tan^2 x &= \frac{1}{3} \\ \tan x &= \pm \sqrt{\frac{1}{3}} = \pm \frac{1}{\sqrt{3}} \\ \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) &= 30^\circ \end{aligned}$$



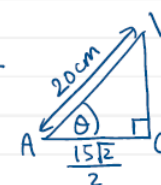
$$\begin{aligned} 4 \quad (a) \quad y &= 4x^2 + 5x + 3 \\ y &= x + 2 \\ x + 2 &= 4x^2 + 5x + 3 \\ 4x^2 + 4x + 1 &= 0 \\ (2x + 1)^2 &= 0 \\ 2x + 1 &= 0 \\ x &= -\frac{1}{2} \\ y &= \left(-\frac{1}{2}\right) + 2 \\ &= \frac{3}{2} \\ \therefore \left(-\frac{1}{2}, \frac{3}{2}\right) \end{aligned}$$

$$\begin{aligned} (b) \quad x^2 + y^2 - 10x + 16y - 80 &= 0 \\ x^2 - 10x + 25 + y^2 + 16y + 64 &= 80 + 25 + 64 \\ (x - 5)^2 + (y + 8)^2 &= 169 \\ \text{Centre} &= (5, -8) \\ \text{radius} &= \sqrt{169} = 13 \end{aligned}$$

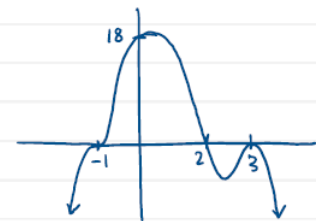


$$\begin{aligned} AC &= \sqrt{15^2 + 15^2} \\ &= 15\sqrt{2} \\ AO &= \frac{15\sqrt{2}}{2} \end{aligned}$$

$$\begin{aligned} \cos \theta &= \frac{15\sqrt{2}}{20} \\ &= \frac{3\sqrt{2}}{8} \\ \theta &= \cos^{-1}\left(\frac{3\sqrt{2}}{8}\right) \\ &= 58^\circ \end{aligned}$$



$$\begin{aligned} (d) \quad y &= -(x+1)^3(x-2)(x-3)^2 \\ y\text{-int: Sub } x=0 \\ y &= -(0+1)^3(0-2)(0-3)^2 \\ &= 18 \\ x\text{-int: Sub } y=0 \\ 0 &= -(x+1)^3(x-2)(x-3)^2 \\ x &= -1, 2, 3 \end{aligned}$$



$$\begin{aligned} 5 \quad (a) \quad \tan x \cos x \operatorname{cosec} x \\ &= \frac{\sin x}{\cos x} \times \cos x \times \frac{1}{\sin x} \\ &= 1 \end{aligned}$$

$$\begin{aligned} (b) \quad (\text{there are multiple methods}) \\ \text{LHS} &= \frac{1 - \sin^2 x \cos^2 x}{\sin^2 x} \\ &= \frac{1 - \sin^2 x (1 - \sin^2 x)}{\sin^2 x} \\ &= \frac{1 - \sin^2 x + \sin^4 x}{\sin^2 x} \\ &= \frac{1 - \sin^2 x}{\sin^2 x} + \frac{\sin^4 x}{\sin^2 x} \\ &= \frac{\cos^2 x}{\sin^2 x} + \sin^2 x \\ &= \cot^2 x + \sin^2 x \\ &= \text{RHS} \end{aligned}$$

$$(c) 2\sin^2 x - \cos^2 x - \frac{1}{2} \sin x = \frac{3}{2}$$

$$2\sin^2 x - (1 - \sin^2 x) - \frac{1}{2} \sin x = \frac{3}{2}$$

$$2\sin^2 x - 1 + \sin^2 x - \frac{1}{2} \sin x = \frac{3}{2}$$

$$3\sin^2 x - \frac{1}{2} \sin x - \frac{5}{2} = 0$$

$$6\sin^2 x - \sin x - 5 = 0$$

$$(6\sin x + 5)(\sin x - 1) = 0$$

$$6\sin x + 5 = 0 \text{ or } \sin x - 1 = 0$$

$$\sin x = -\frac{5}{6} \quad \sin x = 1$$

$$\sin^{-1}\left(-\frac{5}{6}\right) = 56^\circ \quad x = 90^\circ$$



$$x = 180 + 56, 360 - 56$$

$$x = 236^\circ, 304^\circ$$

$$\therefore x = 90^\circ, 236^\circ, 304^\circ$$